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Hiring Committee  
Inria  
France

To the Hiring Committee,

I am writing to express my strong interest in the PhD Thesis position on Speech Enhancement with Machine Learning and Inverse Methods. As an AI & Data Science Engineer with deep expertise in signal processing, Python, and advanced neural network architectures, I am highly motivated to tackle the challenges of acoustics and speech signal enhancement.

My engineering background has consistently bridged complex physical signals with deep learning. In my "Cardiovascular Medical Digital Twin," I engineered **Medical\_MambaTS**, a state space model designed for sub-millisecond, linear-time processing of high-frequency (100Hz) temporal signals. Speech enhancement requires the same rigorous treatment of temporal dependencies and noise robustification. Furthermore, my implementation of a physics-informed DeepONet utilizing PyTorch Autograd demonstrates my ability to integrate inverse problems and physical constraints directly into deep learning architectures—an approach perfectly suited to acoustic wave propagation and inverse methods.

I am eager to apply my strong mathematical foundation and deep learning expertise to push the boundaries of speech enhancement at Inria.

Sincerely,

Ismail Aissaoui